

Completing Short Algebraic and Segment Proofs

- Every line of a proof needs a reason. Use the properties of equality (addition, subtraction, multiplication, division, substitution, distributive), the definitions of midpoint and congruent segments, and the Segment Addition Postulate.
 - All lengths are given with units. Answer with the unit shown on the answer line.
 - Diagrams show labelling only; the measurements are in the problem text.
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1. A shelf support of total length 22 cm is built from three equal spacers of length x cm and one 7 cm bracket, so $3x + 7 = 22$.

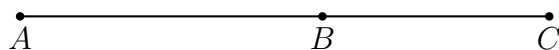
(a) Find the spacer length x .

(b) Complete the reason column of the proof below.

Statement	Reason
1. $3x + 7 = 22$	1. Given
2. $3x = 15$	2.
3. $x = 5$	3.

Answer: _____ cm

- 2.** Three markers stand in a straight line along a trail, in the order A , B , C . The distances, in metres, are $AB = 2x$, $BC = x + 4$, and $AC = 25$.



- (a) Use the Segment Addition Postulate to write an equation, then find x .

Answer: _____

- (b) Find the distance BC .

Answer: _____ m

- 3.** A straight handrail $\overline{PQ}$ has its midpoint M marked. Complete the reason column to prove that PM is half of PQ .

Statement	Reason
1. M is the midpoint of $\overline{PQ}$	1. Given
2. $\overline{PM} \cong \overline{MQ}$	2.
3. $PM = MQ$	3.
4. $PM + MQ = PQ$	4.
5. $PM + PM = PQ$, so $PM = \frac{1}{2}PQ$	5. Substitution and division property

4. Two cable runs on a bracket have lengths $5(x - 3)$ cm and $(2x + 9)$ cm, and they are equal.

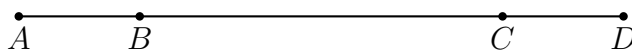
(a) Find x .

(b) Write the reason for each line of the proof below.

Statement	Reason
1. $5(x - 3) = 2x + 9$	1. Given
2. $5x - 15 = 2x + 9$	2.
3. $3x - 15 = 9$	3.
4. $3x = 24$	4.
5. $x = 8$	5.

Answer: _____ cm

5. Four survey markers lie in a straight line along a fence, in the order A, B, C, D . It is given that $AB = CD$.



- (a) Prove that $AC = BD$. Write your argument in two or three sentences, naming a reason for each step.

- (b) The markers are then measured: $AB = CD = 3.5$ m and $BC = 12$ m. Find AC .

Answer: _____ m

6. The same four markers A, B, C, D lie in a straight line, but now it is given that $AC = BD$. Complete the reason column to prove $AB = CD$.

Statement	Reason
1. $AC = BD$	1.
2. $AB + BC = AC$	2.
3. $BC + CD = BD$	3.
4. $AB + BC = BC + CD$	4.
5. $AB = CD$	5.

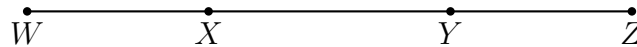
7. A rectangular garden bed has perimeter $P = 2l + 2w$. Its perimeter is 46 m and its width is 8 m.

(a) Find the length l .

Answer: _____ m

(b) Name the property of equality that justifies each step of your work: replacing P and w by their measured values, removing the doubled width from both sides, and the final step.

8. A footbridge rests on four supports W , X , Y , Z set in a straight line in that order. The two outer spans are equal, $WX = YZ$. The middle span is $XY = 6.4$ m and the whole bridge is $WZ = 21.0$ m.



(a) Prove that $WY = XZ$, naming a reason for each step.

(b) Find the length of one outer span WX .

Answer: _____ m